

MANAV RAMPRASAD

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EDUCATION

UNIVERSITY OF TORONTO - TORONTO, ON, CANADA

SEP 2025 - PRESENT

- Ph.D. Student in Computer Science 4.0/4.0
 - Supervised by Dr. Alán Aspuru-Guzik,
 - Working in the Matter Lab, affiliated with the Vector Institute
- Relevant Courses: Quantum Algorithms, ML for Drug Discovery, Deep Learning & Neural Networks
- Teaching Assistant for CSC108: Intro to Programming (x2)

GEORGIA INSTITUTE OF TECHNOLOGY - ATLANTA, GA, USA

JUN 2020 - MAY 2024

- Bachelor of Science in Computer Science - Intelligence & Modeling/Simulation 3.8/4.0
 - Minor in Materials Science & Engineering
- Scholarships/Awards: NSF GRFP Honorable Mention, Zell Miller Scholarship, M. Weldon Rogers Scholarship

EXPERIENCE

MATTER LAB - TORONTO, ON

SEP 2025 - PRESENT

- Leading an LG-funded initiative focused on using quantum algorithms to simulate phase change on a cold plate
- Aiding in development of multi-agent systems designed for quantum computing and theoretical math problems
- Evaluated uncertainty quantification in co-folding models to show drop in performance in out of distribution data

FUNG LAB - ATLANTA, GA

MAY 2024 - MAY 2025

- Pre-trained on atomic-level descriptors including Atom-Centered Symmetry Functions and Embedded-Atom Density
- Fine-tuned on the Materials Benchmark dataset created by the Materials Project to create a new proprietary model
- Feature engineered to find new and more holistic descriptor functions to generate features on for pre-training

CHAO ZHANG RESEARCH GROUP - ATLANTA, GA

JUN 2022 - NOV 2024

- Improved In-Context learning and multi-step reasoning for healthcare texts by training over MIMIC-III notes
- Created PostgreSQL databases to analyze the MIMIC-III dataset, containing over 50,000 distinct hospital visits
- Pre-processed data in the database to recreate existing Machine Learning models' results to track improvement

PUBLICATIONS/PRESENTATIONS

- Gustin et al "El Agente Cuantico: Automating quantum simulations", Pre-Print
- Jia, Govil, Ramprasad, Fung "Pre-training Graph Neural Networks with Structural Fingerprints for Materials Discovery", Pre-Print
- Xu, Xu, Ramprasad, Tumanov, Zhang "TransEHR: Self-Supervised Transformer for Clinical Time Series Data", Machine Learning for Health
- Tran et al "Machine-learning predictions of polymer properties with Polymer Genome", Journal of Applied Physics
- Ramprasad, Kim "Assessing and Improving Machine Learning Model Predictions of Polymer Glass Transition Temperatures", Journal of Emerging Investigators

SKILLS

SOFTWARE:	Python, Qiskit, PyTorch, TensorFlow, Java, PostgreSQL, R, MATLAB, G-Code (CNC), AutoCAD
ACADEMICS:	Competition Math (AIME qualifier, NEML winner), CS (CS Olympiad), Physics (Physics Olympiad)
COMMUNICATION:	Public Speaking & Presentations (WE Day Madison Square Garden), Research Paper Writing
EXTRACURRICULARS:	Chess (GT Team, Former CT State Champion - USCF Rating: 1879), Basketball, AKPsi Fraternity